FORM 2

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

Hybrid Detection–Segmentation for Ground-Penetrating Radar: A Unified Dataset Builder, YOLO–U-Net Architecture, and Radar-Aware Fusion Framework for Robust Subsurface Feature Mapping

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3. PREAMBLE TO THE DESCRIPTION:

The following specification describes the nature of the invention and the manner in which it is to be performed. The invention relates to an automated system and method for analyzing Ground Penetrating Radar (GPR) data by employing a hybrid machine-learning framework that integrates radargram preprocessing, unified dataset generation, deep-learning based object detection, pixel-level segmentation, and result fusion. The system utilizes consolidated annotation formats to train detection and segmentation models simultaneously and combines their outputs to achieve accurate identification, localization, and mapping of subsurface utilities, voids, rebar, and structural anomalies. All variations or equivalent configurations that apply the core inventive principles of hybrid dataset construction, dual-model analysis, and fused subsurface interpretation shall fall within the scope of this invention.

3.1 TECHNICAL FIELD:

The present invention relates to the field of Ground Penetrating Radar (GPR) data analysis, subsurface imaging, and non-destructive underground assessment. More particularly, the invention pertains to intelligent systems and computational methods that employ machine learning, computer vision, and deep neural network architectures for the automated detection, localization, classification, and pixel-level segmentation of underground utilities, structural discontinuities, voids, rebar, conduits, and other hidden subsurface features captured within radargrams. The invention also encompasses techniques to process data before analysis, to harmonize the results and to interpret the overall findings via hybrid model approaches. These techniques are designed to augment the quality, consistency, and validity of ground-penetrating radar (GPR) evaluations in many different types of environments, soil types, and methods of operation.

3.2 PROBLEM ADDRESSED BY THE INVENTION:

There are several ways to interpret GPR data but still requires considerable effort and input from experts with experience using the tool to accurately identify features beneath the surface of the earth. One of the drawbacks of current methods is that they take too long to complete and can result in inconsistencies because they are based upon human observations of hyperbola patterns (alternating reflections as well as signatures) generated during a GPR survey. In addition, existing methods are influenced heavily by noise, may yield inconsistent results depending upon soil type, antenna frequency, and environmental conditions. Methods using current computer vision and signal processing techniques yield minimal accuracy at present; two distinct tasks in most current methods (object detection and segmentation) lead to limited subsurface characterization and high false detection/projection rates. The inability to obtain standardized datasets with pixel-level annotated GPR imagery restricts the ability to develop viable automated Deep Learning models for GPR analysis and therefore increases the time it takes for GPR datasets to be processed. The current invention addresses these deficiencies by providing an automated hybrid detection-segmentation system that operates through a central dataset generation and fusion methodology, which will allow users to accurately and consistently interpret subsurface features based upon processing of raw radargram data with minimal reliance upon expert input.

3.3 SOLUTION TO THE PROBLEM:

An automatic and intelligent method of interpreting GPR data is provided by the invention by means of a hybrid architecture using ML techniques that perform object detection and pixel-level segmentation of subsurface detection features at the same time. The invention provides a complete set of processing options for GPR data to help enhance the radargrams and remove unwanted noise, improve standardization of GPR data and produce a target mask for the next two steps of the processing. A new method for building a hybrid dataset based on bounding box and polygon annotations has been developed to provide uniform datasets to allow for the simultaneous training of FMC and U-Net based detection segmentation networks. The Yolo-based detection model is used to identify and locate hyperbolic signatures and underground structural features in GPR data, while the U-net based segmentation model is used to define the geometric forms of the features and the extent of the feature. By fusing the outputs of these two models through the developed fusion process, it is possible to achieve higher confidence in the detection, lower rates of false positives, and improved clarity of structure in GPR data. Eliminating manual interpretation of GPR data, establishing processes for standard dataset preparation, and providing a dual model learning pipeline process, have greatly improved the accuracy, robustness, scalability, and efficiency of performing subsurface evaluations using GPR data generated by the invention.

3.4 ADVANTAGES OF THE INVENTION:

- **Automated GPR (Ground Penetrating Radar) data interpretation** obviates the necessity for manual interpretation by specialists, diminishing bias potential when analyzing subterranean features.

- **Combining two approaches to detection/segmenting**, one via Spatio-temporal bounding boxes, the other at pixel-level, increases accuracy in the form of better detection, discrimination, interpretation capability.
- The synchronization of the datasets will allow for the accurate generation of a unified dataset for both Detection and Segmentation models. The creation of this **Unified Dataset** should also result in improved learning outcomes and also remove inconsistencies produced by the creation of separate datasets for each type of Model.
- With the ability to produce reliable results in many types of soil environments, including variations in moisture level ("wet" to "dry"), differences in radio frequency (low-frequency versus high-frequency), and the materials and rock types where you find buried infrastructure, this system has become the **more robust survey solution**.
- Fusion of "YOLO" detection result data and "U-Net" segmentation produced **fewer false positives/negatives**.
- This system may process **large amounts of survey data within a very short time** and provide options for nationwide inspection and mapping of buried utilities, as well as provide information regarding subsurface safety.
- Users may better isolate and view the features they are identifying. This results in a **greater ability to identify buried pipes, voids, rebars, cavities, etc.**
- The **ability to operate across multiple industries**, such as civil engineering, archaeological site development, construction site/layout identification, defense applications, mining industry, and geophysics.
- The use of a **modular design** allows for substitute deep learning algorithms and configuration methods, or file formats for data, while preserving the fundamental inventor's method.
- The **significant cost and time savings** associated with subsurface surveys created by the ability to use automated detection systems that can be implemented on laptops, embedded systems and cloud systems.
- **Data obtained is more diverse**, allowing for improved robustness of trained networks through the creation of segmentation masks and the pairing of bounding boxes.

4. DESCRIPTION:

4.1. Field of invention:

The new technology for Ground Penetrating Radar (GPR) to explore underground environments and interpret results uses several cutting-edge technologies including Artificial Intelligence (Machine Learning (ML)), computer vision and Deep Neural Network (DNN) algorithms. It automates the detection, location and segmentation of sub-surface objects and structures such as pipes and wiring not visible by traditional means from radar data. Also included are a method for pre-processing radar data before it is evaluated and a method for creating consistent annotation or labelling conventions from different analysts' lists of findings regarding radar images. Furthermore, using multiple new hybrid analytical methods will result in greater precision, accuracy and efficiency when conducting non-destructive testing of GPR and underground mapping with this advanced technology.

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4.2. Background of Invention:

- Ground Penetrating Radar (GPR): This is a non-destructive approach to subsurface imaging that has been extensively applied in civil engineering, structural, utility mapping, archaeology and geophysical exploration.
- GPR radargrams interpretation is mainly by hand examination of the hyperbolic reflections and signal patterns, through a trained expert, and is therefore time-consuming, tedious and subject to human error.
- Differences in soil structure, moisture content, antenna frequency, layers of the geology, and external interference normally cause variation of radar signatures making the accurate interpretation of the visuals more difficult.
- The image-processor and signal-processor methods of GPR analysis provide a low degree of flexibility and accuracy, particularly when operating within a noisy or intricate setting.
- The current research AI-based methods are either detection-based or segmentation-based, and they do not provide deep structural insights that could underlie the reliability of the underground feature classification.
- There are inconsistencies in the synchronized datasets for bounding-box labels and the pixel-level segmentation masks. Due to the lack of a defined and properly organized dataset containing all of

the bounding-box labels in the same format (synchronized) and pixel-level segmentation masks in the same format, there is a lack of resources for building deep learning algorithms that can utilize both location information, as well as structure-oriented geometry.

- Current workflows do not offer a unified structure, which can preprocess radargrams, reconcile multi-format annotations, and train dual-purpose models and integrate results to produce strong and automated subsurface analysis.
- In line with this, it is evident that there is a demand to have an advanced scalable and automated solution to incorporate preprocessing, hybrid dataset generation, detection, segmentation and fusion techniques to precisely identify and map the buried utilities and structural abnormalities with minimum input of the expert.

4.3. Brief Description of Drawing/ Figures:

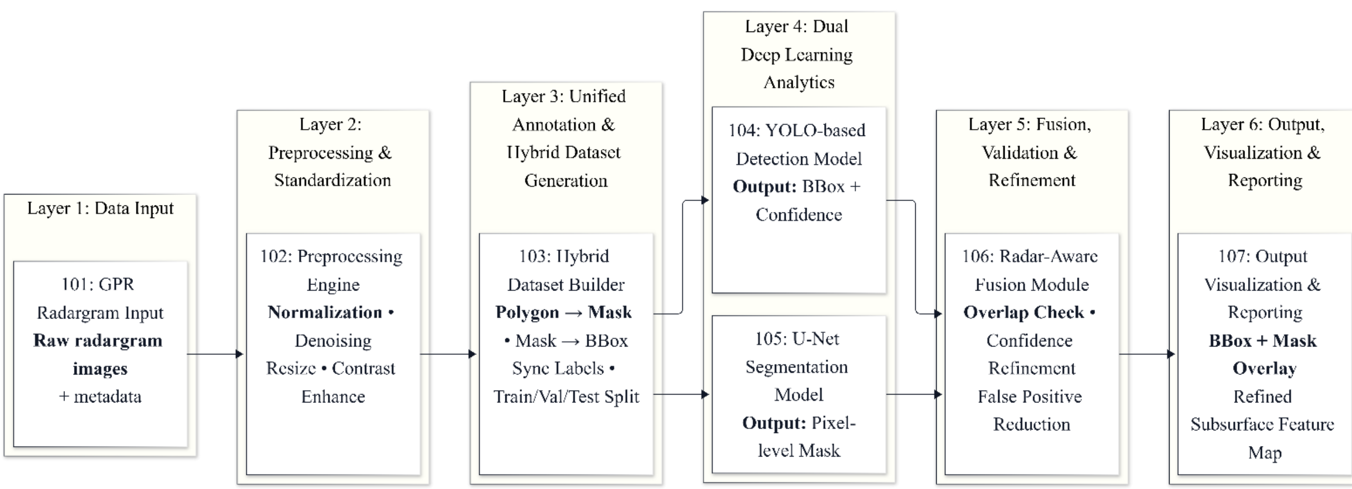

Fig 1. System Architecture

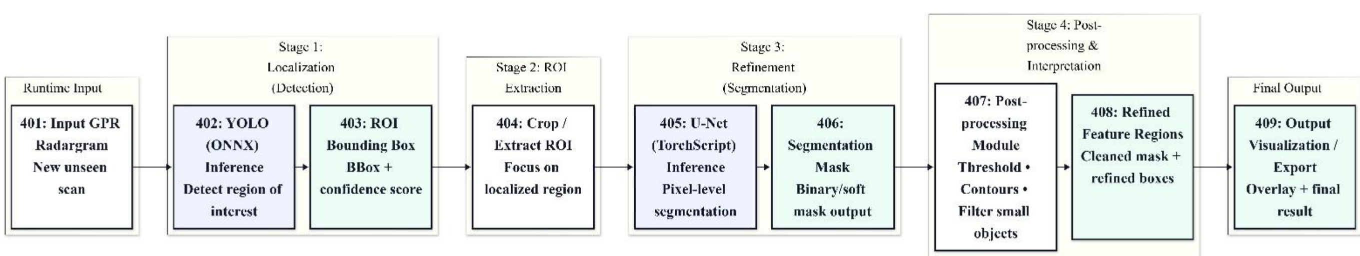

Fig 2. Inference Pipeline

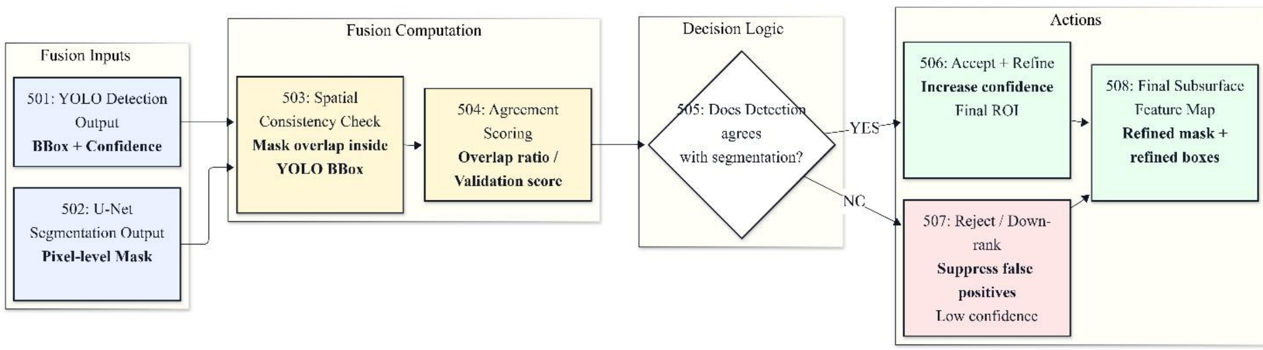

Fig 3. Fusion Logic Diagram

4.4. Detailed Description of Development Process:

- **Modular System Architecture:**
  The invention was formulated as a framework in a modular way; it has five fundamental units, such as Preprocessing Engine, Hybrid Dataset Builder, Detection Model, Segmentation Model and Fusion Module.

- **Preprocessing Engine:**
  Written to improve the raw GPR radargrams with the help of the normalization, contrast optimization, resizing protocols, and noise reduction filters. The engine also produced binary masks of polygonal annotations of segmentation tasks.

- **Hybrid Dataset Builder:**
  Introduced to standardize various annotation formats, by deriving bounding-box labels, rasterizing polygon contours and aligning the two into a common dataset format. This builder created multi-purpose training files to be used in the detection and segmentation and proper hierarchy of directories.

- **Detection Model (YOLO):**
  Learned on the harmonized dataset to find underground structures by detecting hyperbolic patterns

and structural aberrations. The model configurations, the dimensions of the anchors and training cycles were optimized in such a way that they fit typical GPR signal patterns.

- **Segmentation Model (U-Net):**
  Created to obtain the full geometry of features that have been identified by producing pixel-based masks. Skip connections, encoder-decoder networks were optimized to be more precise in radargram segmentation.

- **Fusion Module:**
  Applied to merge the bounding-box predictions of YOLO with pixel mask of U-Net to permit a higher level of confidence, lowering false positives, and a better isolation of features in the radargram.

- **Backend Analyzer:**
  Designed to evaluate model outputs, generate performance metrics, measure detection confidence, and validate segmentation quality across multiple soil and frequency variations.

- **Visualization Front-End:**
  Constructed to display bounding boxes, segmentation overlays, and fused subsurface renderings for clear interpretation of detected underground utilities.

- **Communication & Synchronization:**
  All modules were integrated through standardized data exchange formats, ensuring interoperability between preprocessing workflows, model outputs, and result aggregation components.

- **Validation & Testing:**
  The complete system was tested on hybrid datasets containing annotated GPR scans, benchmarking accuracy, robustness, and false detection rates. Iterative retraining, tuning, and augmentation cycles were used to improve model performance.

4.5. Detailed description of the invention:

The present invention describes a novel automated process to interpret Ground Penetrating Radar (GPR) data through a combination of two types of Artificial Intelligence (AI): Hybrid Intelligence and Supervised

Learning. An automated system is created to provide both object detection capabilities as well as pixel-level segmentation of subsurface features.

A Preprocessing Engine is utilized in this as it improves raw radargrams by eliminating noise and normalizing, resizing, and masking them. Additionally, the Preprocessing Engine converts (1) polygon annotations from radargrams to a common format; (2) bounding boxes provide a unified representation of the data collected using multiple devices.

Using the Hybrid Dataset Builder, a structured, harmonized dataset is generated by organizing the previously annotated radargram images into single harmonized datasets of similar characteristics and containing all identified anomalies, with the ability to train both detections and segmentations simultaneously. With the help of a YOLO-based model to perform object detections and locate hyperbolic reflections and structures, the U-Net-based model provides pixel-perfect geometric layouts and itemized boundaries related to buried utilities, void spaces and other subsurface features.

SUMMARY:

The invention provides an automated system for analysing Ground Penetrating Radar (GPR) data using a hybrid deep-learning approach that combines object detection and pixel-level segmentation. The system preprocesses radargrams, converts annotations into unified formats, and builds synchronized datasets to train a YOLO-based detection model and a U-Net-based segmentation model. Their outputs are merged through a fusion mechanism that enhances accuracy, reduces false positives, and produces reliable subsurface maps showing the location, shape, and extent of buried utilities and structural anomalies. The invention improves GPR interpretation by eliminating manual dependency, increasing robustness across environmental variations, and enabling scalable, high-confidence underground assessment.

5. CLAIMS

We claim :

a. An automated GPR-based subsurface interpretation system comprising modules for radargram preprocessing, unified dataset construction, detection model execution, segmentation model execution, and output fusion.
b. The preprocessing engine that is set up to improve the raw GPR scan by performing normalization, contrast adjusted, noise-reduced, geometrically resized and changing polygon annotations into segmentation masks.
c. A hybrid dataset builder adapted to synchronize image files, bounding-box labels, and pixel- level masks into unified training structures for combined detection and segmentation learning.
d. A detection model, preferably of a deep-learning type such as YOLO, configured to identify and localize subsurface objects and hyperbolic signatures within radargram data.
e. A segmentation model, preferably U-Net or equivalent architecture, configured to extract pixel-level geometry and structural boundaries of utilities, voids, rebar, conduits, and other subsurface features.
f. A fusion mechanism that merges detection outputs with segmentation masks to reduce false positives, improve feature clarity, and generate refined subsurface interpretations.
g. A method of automated GPR analysis comprising preprocessing of radargrams, constructing synchronized datasets, training detection and segmentation networks, and fusing their outputs to map underground structures.
h. A system as claimed above wherein the models operate sequentially or in parallel depending on computational resource availability.
i. A non-transitory computer-readable medium storing instructions for performing the method of automated subsurface feature analysis as described in the foregoing claims.

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6. ABSTRACT OF THE INVENTION:

The invention provides an automated system and method for interpreting Ground Penetrating Radar (GPR) data through a hybrid deep-learning architecture that performs simultaneous detection and segmentation of subsurface features. The system preprocesses radargrams via normalization, enhancement, resizing, and conversion of polygonal annotations into binary masks, followed by construction of unified datasets containing synchronized bounding-box labels and segmentation targets. A detection network, preferably based on YOLO, identifies and localizes underground utilities, voids, reinforcement elements, and structural anomalies, while a segmentation network, preferably U-Net, extracts pixel-accurate geometries of these features. The results of the two models are further combined to remove any false detections, refine the boundaries and give an accurate interpreted subsurface map. The invention removes the need to perform manual interpretation, enhances the level of reliability in the analysis of GPR data in conditions of variable soil and signal, and offers a scalable framework of automated structural assessment and mapping the underground.